

Competing wireless communications technologies

Standard	Family	Application	Multiplexing	Notes
802.16e	WiMAX	MobileInternet	SOFDMA	Quoted speeds achievable at short range; practically 10Mbps at 10Km
HIPERMAN	HIPERMAN	MobileInternet	OFDM	
WiBro	WiBro	MobileInternet	OFDM	Short range (<5km)
UMTS W-CDMA HSDPA+HSUPA	UMTS/3GSM	Mobile phone	CDMA/FDD	Roadmap shows HSDPA up to 28.8Mbps from the base station. Users can get D/L of 400-600 Kbps and ~100Kbps uplink
UMTS-TDD	UMTS/3GSM	Mobile Internet	CDMA/TDD	Reported speeds according to IPWireless

exit barriers for themselves. These players would find it economically unviable to shift to another technology such as WiMAX without having first gathered the fruits of their investments. But in India, with the delay in implementation of 3G, both WiMAX and 3G could start the race for market penetration on equal footing.

■ **Need:** in a country like India where the penetration of broadband and dial-up has been limited due to the weak physical infrastructure, the need for newer technologies that could reduce costs and dependence on physical infrastructure, is even greater. While 3G is a strong technology, improving mobile communication standards substantially, it still trails behind advantages such as lower cost of last mile connectivity, limited dependence on physical infrastructure, faster networks, higher data hauling capabilities, limited data fade, and enhanced mobility, offered by WiMAX.

■ **Availability of supporting applications:** In India most consumers still use phones, computers, laptops and other devices which do not support 3G or WiMAX. Due to talk of 3G being launched in India having been going on for some years, there are a number of 3G supporting phones now available in here, while the devices supporting WiMAX are yet to be launched in India. With big players such as Nokia and IMB, supporting WiMAX in a big way and their plans for launching chips and gadgets supporting WiMAX in the near future, this issue will surely be taken care of.

■ **Cost of migration for consumers:** As discussed, WiMAX is a relatively inexpensive technology being targeted for the masses by the big players. In the case of WiMAX, interfacing equipment is available helping old consumers shift to the new technology with their old hardware, and the cost for the tran-

sition is expected to be very low due to the low cost of related technology. Also, the cost of new hardware supporting WiMAX would be similar to that of their non-WiMAX versions. The same does not apply to 3G where most mobile phones supporting it are substantially expensive and require complete discard of old handsets.

■ **Telecom player's perspective:**

● **Fixed line:** 3G is a threat for fixed line operators as it provides enhanced capabilities to the mobile operators, providing an opportunity for them to attract more consumers and take more market share away from fixed line. WiMAX provides an opportunity for fixed line players to improve their competitiveness through reduction of costs around the physical infrastructure, and improved performance capabilities by shifting to VoIP.

● **Mobile:** Mobile operators worldwide have recognized the importance of 3G, yet WiMAX would remain attractive, owing to its enhanced data transmission capabilities, providing an opportunity to offer enhanced data and communication services.

■ **Applications:** as discussed earlier, both 3G and WiMAX would support the evolution of enhanced communication products and services, but the impact of WiMAX is expected to be bigger and goes beyond cellular communication — affecting a wide range of services including cable TV, fixed line telecommunication, network data backhaul, etc.

Close scrutiny suggests that both 3G and

Specifications for 3G and WiMAX

Parameter	3G	WiMAX
Communication for stationary devices	Download 2 Mbps; Upload 0.15 Mbps	Download 10 Mbps; Upload 2-3 Mbps
Communication for mobile devices	Low speed 0.14 Mbps High latency (>100 ms)	High speed 2 Mbps Low latency (<100 ms)
Bandwidth	Theoretical maximum 2.4 Mbps	Theoretical maximum 70 Mbps
Security	High	High
QoS	High	High; latency as low as <10ms

WiMAX have a lot to offer to the Indian market, with WiMAX being a bit more suited to the India condition especially with the delayed deployment of 3G and weak fixed wire & broadband infrastructure. Opportunities are vast for 3G and WiMAX in India with the possibility of them co-existing as complementary technologies in the wake of strong communication and coverage needs here. It is likely that both technologies would continue to operate in India with the market being fragmented between these two new technologies and other legacy systems. ■

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